

Addressing Statistical Heterogeneity in Federated Learning For Sea Ship Datasets

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Introduction

Data is critical for training a machine learning model, however creating a robust model without sacrificing personal data is a formidable challenge. The Federated Learning paradigm was created to protect users' data and train a collaborative model that incorporates knowledge without compromising privacy. The challenge however, is in real-time, the models are trained using heterogeneous data, which can hamper the accuracy of the federated model (see Fig 1).

Background Information

Heterogeneity stems from skewed labels, skewed feature distribution, similar features with different labels, and discrepancies in granularity of labeling strategy. We aim to

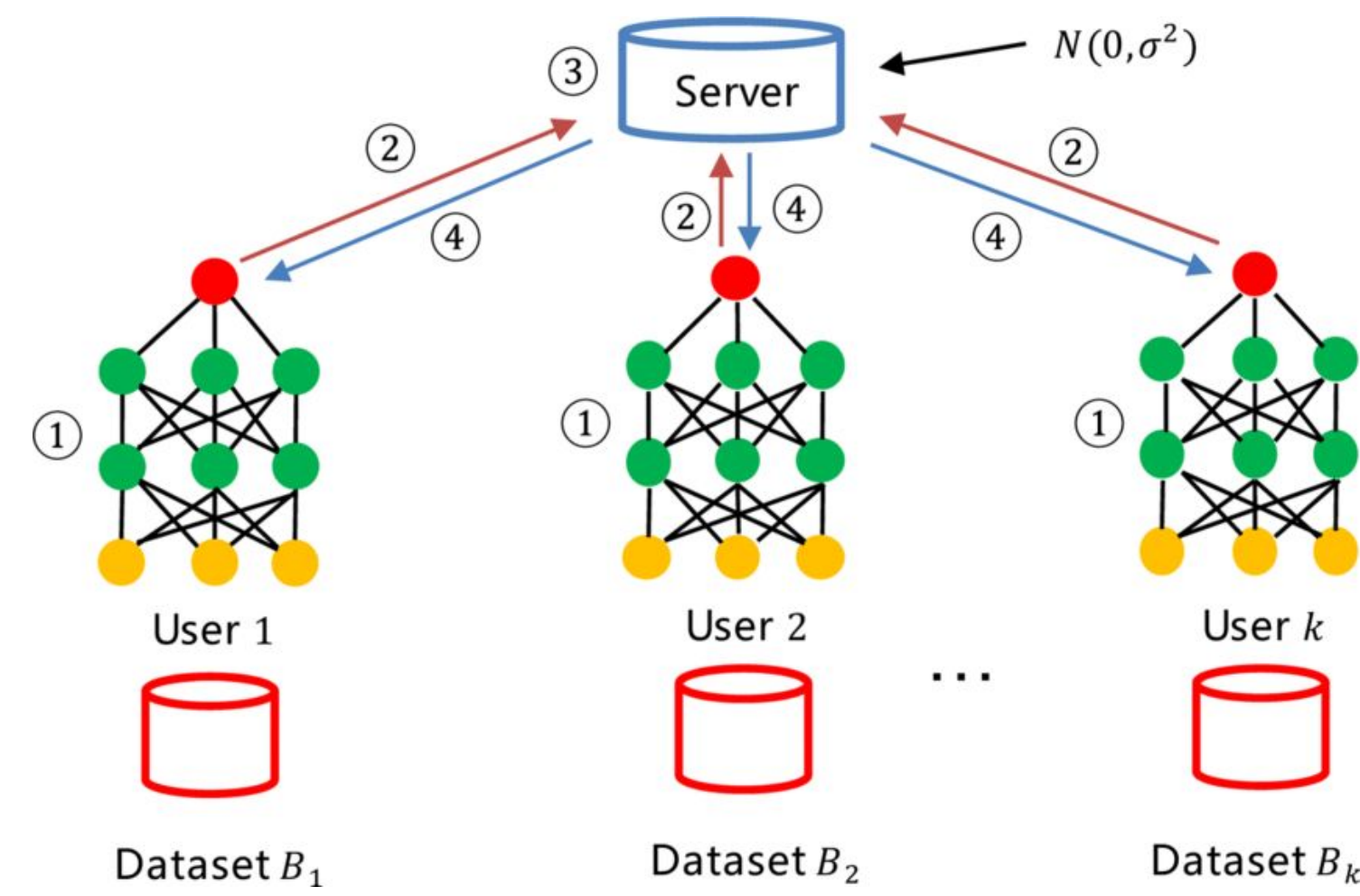


Fig 1: Federated Learning Architecture [5]

converge the heterogeneity and use the technique to understand how the data is distributed across clients. Related work addresses the statistical heterogeneity by focusing on the model weights to optimize convergence [2] Other approaches involve tracking and adapting to client

patterns to reduce bias during the training process due to resource heterogeneity [3] and clustering the clients themselves on specialized local aggregation servers. [4].

Statement of the Research Problem

We propose a method to homogenize heterogeneous datasets for training a federated learning model and determining necessary granularity for accurate model performance.

Data/Results/Analysis

- Datasets: ABOships, SeaShips, Singapore Maritime VIS Onshore and Offshore
- Cluster classes based on feature distribution
- Homogenized multiple datasets which can be applied across clients in a federated learning setting

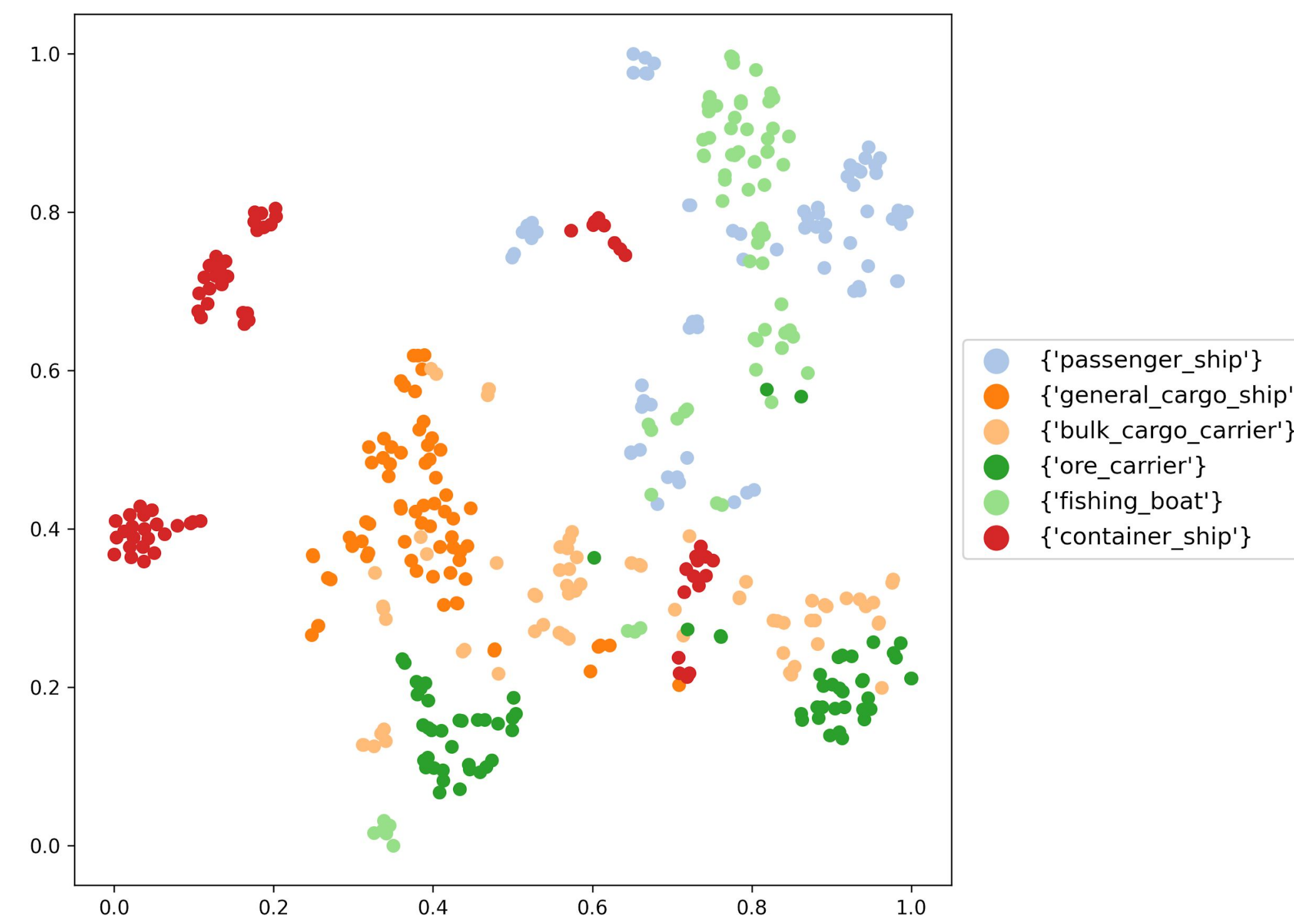


Fig 2: T-SNE Visualization of Clustered Classes for SeaShip Dataset

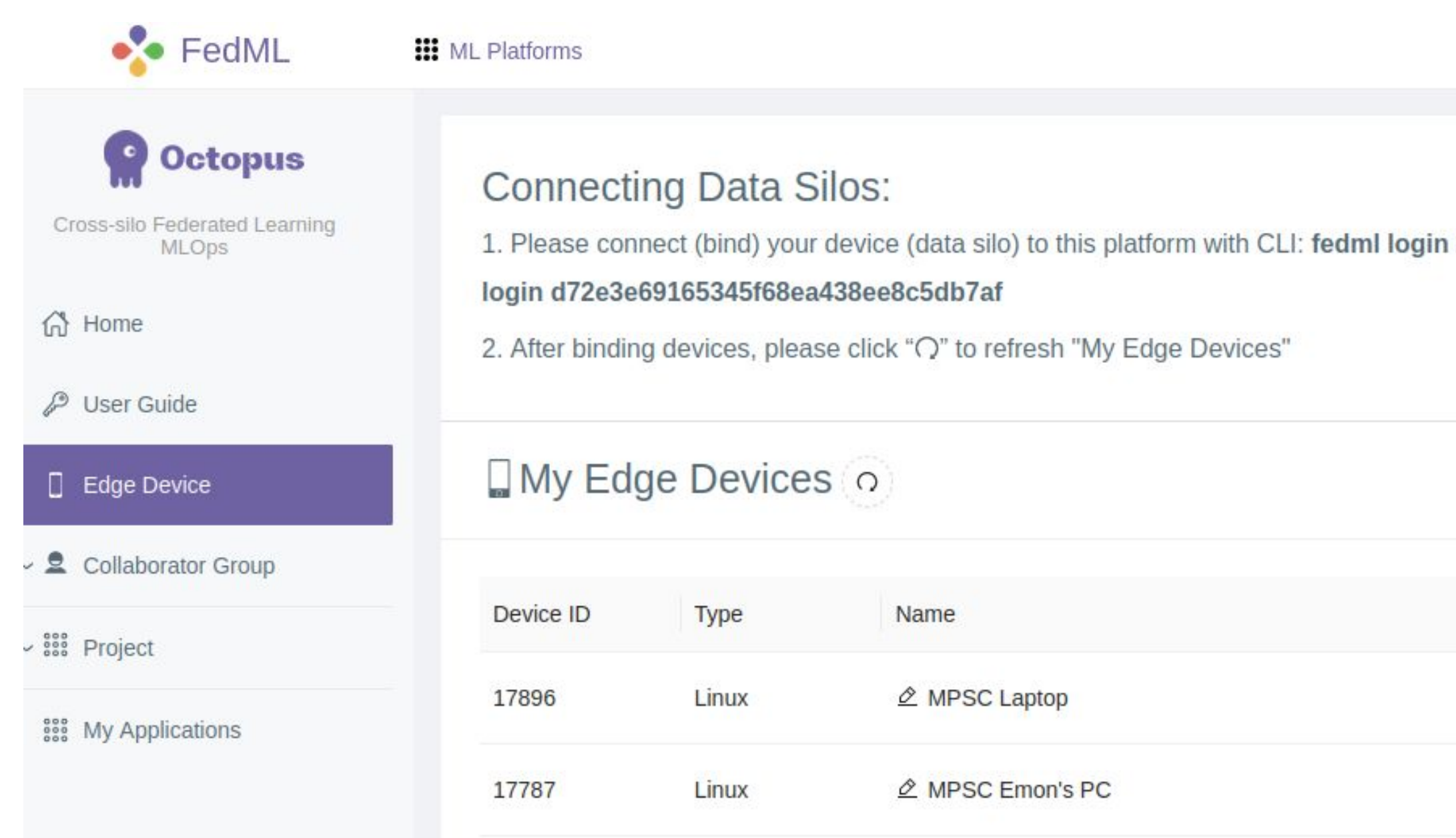


Fig 3: FedML - ML OPS Configuration

Methods/Formulas UMBC

We configured the FedML platform to aggregate the federated model using multiple clients. Each client represents an opensource dataset to detect and classify ships. We crop the images using annotations provided by the datasets, run an off the shelf resnet feature extractor [1] to extract features, that we visualize using PCA clustering and high dimensional TSNE plot in Fig 2.

Discussion

We were able to achieve statistical heterogeneity through macro bucket image clustering. This method can be applied to dataset homogenization where there are annotation errors across many types of image datasets. It can also be used for clustering client data across the federated learning framework.

References/Citations

- [1] img2vec (2023) christiansfka [https://tinyurl.com/ccx8w3dv]
- [2] A Andan, *Robust federated learning under statistical heterogeneity via Hessianspectral decomposition*
- [3] J. Han, *Heterogeneity-Aware Adaptive Federated Learning Scheduling*
- [4] B. Gong, *Adaptive Clustered Federated Learning for Heterogeneous Data in Edge Computing*
- [5] Y. Ding, "Federated learning framework with differential privacy update," July 2020, *Research Gate*

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